Quadratic Equation: Formula Sheet

1. Roots and nature

Roots of $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

Discriminant: $D = b^2 - 4ac$

$D > 0$: real, distinct; $D = 0$: real, equal; $D < 0$: imaginary

Perfect square polynomial: $a > 0,\ D = 0$

2. Sum, product, difference

Sum:

$$\alpha + \beta = -\frac{b}{a}$$

Product:

$$\alpha\beta = \frac{c}{a}$$

Difference:

$$|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$$

Cubic $ax^3 + bx^2 + cx + d = 0$:

$$\sum \alpha = -\frac{b}{a},\quad \sum \alpha\beta = \frac{c}{a},\quad \alpha\beta\gamma = -\frac{d}{a}$$

3. Symmetric functions

Squares:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

Cubes:

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

Fourth powers:

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

Reciprocals:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

Ratios:

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

Difference squared:

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

With $t = x + \frac{1}{x}$:

$$x^2 + \frac{1}{x^2} = t^2 - 2,\quad x^3 + \frac{1}{x^3} = t^3 - 3t$$

Complex conjugate roots, $r = \sqrt{\alpha\beta}$:

$$\alpha^n + \beta^n = 2r^n \cos n\theta,\quad \cos\theta = \frac{\alpha + \beta}{2r}$$

4. Formation of equations

Quadratic:

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Cubic: $x^3 - S_1x^2 + S_2x - S_3 = 0$

Factor form: $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

Roots $k\alpha$:

$$x \to \frac{y}{k}$$

Roots $\alpha + k$: $x \to y - k$

Roots $\frac{1}{\alpha}$:

$$x \to \frac{1}{y}$$

5. Newton's formula

$S_n = p\alpha^n + q\beta^n$: $aS_{n+2} + bS_{n+1} + cS_n = 0$

6. Special cases

Exactly one root 0: $c = 0,\ b \neq 0$

Both roots 0: $b = c = 0$

Exactly one root ∞: $a = 0,\ b \neq 0$

Both roots ∞: $a = b = 0,\ c \neq 0$

Identity: $a = b = c = 0$

$x = 1$ is a root: $a + b + c = 0$

$x = -1$ is a root: $a - b + c = 0$

7. Common roots

Both common:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Exactly one common:

$$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$$

8. Graph, sign and range

Vertex:

$$\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$$

Range, $a > 0$:

$$\left[-\frac{D}{4a}, \infty\right)$$

Range, $a < 0$:

$$\left(-\infty, -\frac{D}{4a}\right]$$

$f(x) > 0\ \forall x$: $a > 0,\ D < 0$

$f(x) < 0\ \forall x$: $a < 0,\ D < 0$

$f(p) = f(q),\ p \neq q$:

$$x_{\text{axis}} = \frac{p + q}{2}$$

$\dfrac{ax+b}{cx+d}$, range:

$$\mathbb{R} - \left\{\frac{a}{c}\right\}$$

9. Location of roots

Both $> d$:

$$D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} > d$$

Both $< d$:

$$D \geq 0,\ af(d) > 0,\ -\frac{b}{2a} < d$$

Both in (d, e):

$$D \geq 0,\ af(d) > 0,\ af(e) > 0,\ d < -\frac{b}{2a} < e$$

d between roots: $af(d) < 0$

One $< d$, other $> e$: $af(d) < 0,\ af(e) < 0$

Exactly one in (d, e): $f(d)f(e) < 0$

One +ve, one −ve: $af(0) < 0$

Both +ve:

$$D \geq 0,\ -\frac{b}{2a} > 0,\ af(0) > 0$$

Both −ve:

$$D \geq 0,\ -\frac{b}{2a} < 0,\ af(0) > 0$$

10. Higher degree and polynomials

S_k for $a_0x^n + \cdots + a_n = 0$:

$$S_k = (-1)^k \frac{a_k}{a_0}$$

Sum of squares of roots: $\sum \alpha_i^2 = S_1^2 - 2S_2$

Remainder theorem:

$$f(x) \div (x-a) \ \Rightarrow\ R = f(a)$$

$a^3+b^3+c^3-3abc$: $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

11. Inequalities

Modulus:

$$|q| < k \iff -k < q < k; \qquad |q| > k \iff q < -k \text{ or } q > k$$

Solution set inside another:

$$[r_1, r_2] \subseteq [s_1, s_2] \iff r_1 \geq s_1,\ r_2 \leq s_2$$

12. Two variables

General form: $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

Two linear factors: $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$ax^2 + 2hxy + by^2 > 0\ \forall (x, y) \neq (0, 0)$: $a > 0,\ h^2 < ab$

13. Substitution ranges

Reciprocal-type equation, $+bx$:

$$t = x + \frac{1}{x} :\ at^2 + bt + (c - 2a) = 0$$

Reciprocal-type equation, $-bx$:

$$t = x - \frac{1}{x} :\ at^2 + bt + (c + 2a) = 0$$

Allowed t:

$$x + \frac{1}{x} :\ |t| \geq 2; \quad x^2 :\ t \geq 0; \quad e^x,\ a^x :\ t > 0$$

Homogeneous in u, v: put $t = \dfrac{u}{v}$, solve in t, then $u = tv$

Four linear factors: pair them so constants have equal sum; common quadratic $= t$